

The Problem

There is a boundary matrix given

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
0	0	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0
1	0	0	0	0	0	0	0	0	0	0	1	1	0	0	0	0
2	0	0	0	0	0	0	0	1	0	0	1	0	1	0	0	0
3	0	0	0	0	0	0	1	0	0	1	0	0	0	0	0	0
4	0	0	0	0	0	0	0	1	1	1	0	0	0	0	0	0
5	0	0	0	0	0	0	1	0	1	0	0	1	1	0	0	0
6	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1
7	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0
8	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1
9	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1
10	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0
11	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0
12	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	0

and we are making **birth-birth switch forward** transposition (2,3).

a	b	x	y
2	10	3	8

Computation Results

Birth Death Pairs

Before the transposition	{(0, 1), (3, 8), (2, 10), (12, 13), (9, 15), (5, 6), (11, 14), (4, 7)}
After the transposition	{(0, 1), (12, 13), (3, 10), (9, 15), (5, 6), (11, 14), (4, 7), (2, 8)}

Ralations

	Algorithm 1	Algorithm 2
Before the transposition	{(13, 14), (6, 8), (7, 8)}	{(5, 3), (3, 2), (4, 2)}
After the transposition	{(13, 14), (6, 8), (8, 10), (7, 8)}	{(2, 3), (5, 3), (4, 2)}

Computed sets

Successors and Predecessors

	$\text{Set}^{\text{bt}}(2, 10)$	$\text{Set}^{\text{bt}}(3, 8)$	$\text{Set}^{\text{at}}(2, 8)$	$\text{Set}^{\text{at}}(3, 10)$
Succ_1	$\text{Succ}_1^{\text{bt}}(2, 10) = \emptyset$	$\text{Succ}_1^{\text{bt}}(3, 8) = \emptyset$	$\text{Succ}_1^{\text{at}}(2, 8) = \{(3, 10)\}$	$\text{Succ}_1^{\text{at}}(3, 10) = \emptyset$
Pred_1	$\text{Pred}_1^{\text{bt}}(2, 10) = \emptyset$	$\text{Pred}_1^{\text{bt}}(3, 8) = \{(5, 6), (4, 7)\}$	$\text{Pred}_1^{\text{at}}(2, 8) = \{(5, 6), (4, 7)\}$	$\text{Pred}_1^{\text{at}}(3, 10) = \{(2, 8)\}$
Succ_2	$\text{Succ}_2^{\text{bt}}(2, 10) = \emptyset$	$\text{Succ}_2^{\text{bt}}(3, 8) = \{(2, 10)\}$	$\text{Succ}_2^{\text{at}}(2, 8) = \{(3, 10)\}$	$\text{Succ}_2^{\text{at}}(3, 10) = \emptyset$
Pred_2	$\text{Pred}_2^{\text{bt}}(2, 10) = \{(3, 8), (4, 7)\}$	$\text{Pred}_2^{\text{bt}}(3, 8) = \{(5, 6)\}$	$\text{Pred}_2^{\text{at}}(2, 8) = \{(4, 7)\}$	$\text{Pred}_2^{\text{at}}(3, 10) = \{(5, 6), (2, 8)\}$

$\mathcal{L}$ and $\mathcal{M}$ sets

	Before the Transposition	After the Transposition
$\mathcal{L} = \{(s, t) \in \text{BD} : f(t) < f(y)\}$	$\{(0, 1), (5, 6), (4, 7)\}$	$\{(0, 1), (5, 6), (4, 7)\}$
$\mathcal{M} = \{(s, t) \in \text{BD} : f(y) < f(t) < f(b)\}$	$\emptyset$	$\emptyset$

Equations

eq	left (formula)	left (para- matrized)	left (value)	right (formula)	right (paramatrized)	right (value)	correct
8	$\text{Succ}_1^{\text{at}}(a, y)$	$\text{Succ}_1^{\text{at}}(2, 8)$	$\{(3, 10)\}$	$\text{Succ}_1^{\text{bt}}(x, y) \oplus \{(x, b)\} \oplus \text{Succ}_1^{\text{bt}}(a, b)$	$\text{Succ}_1^{\text{bt}}(3, 8) \oplus \{(3, 10)\} \oplus \text{Succ}_1^{\text{bt}}(2, 10)$	$\{(3, 10)\}$	✓
9	$\text{Succ}_1^{\text{at}}(x, b)$	$\text{Succ}_1^{\text{at}}(3, 10)$	$\emptyset$	$\text{Succ}_1^{\text{bt}}(a, b)$	$\text{Succ}_1^{\text{bt}}(2, 10)$	$\emptyset$	✓
10	$\text{Pred}_1^{\text{at}}(a, y)$	$\text{Pred}_1^{\text{at}}(2, 8)$	$\{(5, 6), (4, 7)\}$	$\text{Pred}_1^{\text{bt}}(x, y)$	$\text{Pred}_1^{\text{bt}}(3, 8)$	$\{(5, 6), (4, 7)\}$	✓
11	$\text{Pred}_1^{\text{at}}(x, b)$	$\text{Pred}_1^{\text{at}}(3, 10)$	$\{(2, 8)\}$	$\text{Pred}_1^{\text{bt}}(a, b) \oplus \{(a, y)\}$	$\text{Pred}_1^{\text{bt}}(2, 10) \oplus \{(2, 8)\}$	$\{(2, 8)\}$	✓
12	$\text{Succ}_2^{\text{at}}(a, y)$	$\text{Succ}_2^{\text{at}}(2, 8)$	$\{(3, 10)\}$	$\text{Succ}_2^{\text{bt}}(x, y) \oplus \{(x, b), (a, b)\}$	$\text{Succ}_2^{\text{bt}}(3, 8) \oplus \{(3, 10), (2, 10)\}$	$\{(3, 10)\}$	✓
13	$\text{Succ}_2^{\text{at}}(x, b)$	$\text{Succ}_2^{\text{at}}(3, 10)$	$\emptyset$	$\text{Succ}_2^{\text{bt}}(a, b)$	$\text{Succ}_2^{\text{bt}}(2, 10)$	$\emptyset$	✓
14	$\text{Pred}_2^{\text{at}}(a, y)$	$\text{Pred}_2^{\text{at}}(2, 8)$	$\{(4, 7)\}$	$[\text{Pred}_2^{\text{bt}}(a, b) \cap \mathcal{L}]$	$[\text{Pred}_2^{\text{bt}}(2, 10) \cap \mathcal{L}]$	$\{(4, 7)\}$	✓
15	$\text{Pred}_2^{\text{at}}(x, b)$	$\text{Pred}_2^{\text{at}}(3, 10)$	$\{(5, 6), (2, 8)\}$	$[\text{Pred}_2^{\text{bt}}(x, y) \oplus \{(a, y)\} \oplus [\text{Pred}_2^{\text{bt}}(a, b) \cap \mathcal{M}]]$	$[\text{Pred}_2^{\text{bt}}(3, 8) \oplus \{(2, 8)\} \oplus [\text{Pred}_2^{\text{bt}}(2, 10) \cap \mathcal{M}]]$	$\{(5, 6), (2, 8)\}$	✓